

## Theorem: (Existence of least upper bound)

Let  $E$  be a non empty subset of  $\mathbb{R}$  with upper bound  $M$ . We know that  $E$  has at most one least upper bound, we have to show that  $E$  has at least one least upper bound.

Since  $E$  is non empty, we have chosen some element  $x_0$  in  $E$ .

Let  $n \geq 1$  be a positive integer. We know that  $E$  has an upper bound  $M$ .

By Archimedean property.  
we can find an integer  
 $K$  such that  $K/n \geq M$   
and hence  $K/n$  is also  
an upper bound of  $E$ . By  
same Archimedean property  
 $\exists$  another integer  $L$  s.t

$$L/n < x_0.$$

Since  $x_0 \in E$  we see that  
 $L/n$  is not an upper bound  
of  $E$ . Since  $K/n$  is an  
upper bound but  $L/n$  is not  
an upper bound  $\Rightarrow K \geq L$

Since  $K/n$  is an upper bound  $L/n$  is not an upper bound then we can find an integer  $L < m_n \leq K$  with the property that  $m_n/n$  is an upper bound for  $E$  but  $(m_n-1)/n$  is not.

In fact, the integer  $m_n$  is unique ( $m_n \rightarrow m$  is depends on the choice of  $n$ )

$m_1, m_2, \dots, m_n \rightarrow$  seq<sup>n</sup> of integers with each  $m_n/n$  is upper bound and each of

$(m_n - 1)/n$  not being the upper bounds.

Now, let  $N \geq 4$  be a positive integer, let  $n, n' \geq N$

$\therefore m_n$  is u.b.  $\frac{m_{n'} - 1}{n'}$  is not  
u.b.  $\Rightarrow$

$$m_n/n > \frac{m_{n'} - 1}{n'}$$

$$\frac{m_n}{n} - \frac{m_{n'}}{n} > -\frac{1}{n'} \geq -\frac{1}{N}$$

$\rightarrow (1)$

Similarly,

$$\frac{m_n}{n} - \frac{m_{n'}}{n} \leq \frac{1}{n} \leq \frac{1}{N}$$

$\rightarrow (2)$

$$\left| \frac{m_n}{n} - \frac{m_{n'}}{n'} \right| \leq \frac{1}{N}$$

$$\forall n, n' \geq N \geq 1.$$

$\Rightarrow \frac{m_n}{n}$  is a Cauchy sequence.  $\because \frac{m_n}{n}$  are rational numbers, we can now define the real number  $S$  as

$$S = \lim_{n \rightarrow \infty} \frac{m_n}{n}.$$

We can show that

$$S = \lim_{n \rightarrow \infty} \frac{m_n - 1}{n}$$

To conclude we need to show that  $S$  is the least u.b of  $E$

First  $\rightarrow$  it is an upper bound. let  $x$  be an element of  $E$

$\therefore m_n/n$  is u.b of  $E$

$$x \leq m_n/n \quad \forall n \geq 1$$

$$x \leq \lim_{n \rightarrow \infty} m_n/n = S$$

$\Rightarrow S$  is an u.b of  $E$ .

[Now we have to show that  $S$  is least upper

bound]

suppose  $y$  is an u.b.  
of  $E$   $\therefore (m_n - 1)/n$  is  
not an u.b.  $\Rightarrow$

$$y < \frac{m_n - 1}{n} \quad \forall n \geq 1$$

$$y < \lim_{n \rightarrow \infty} (m_n - 1)/n = S.$$

Thus upper bound  $S$  is  
less than or equal to  
every upper bound of  $E$ .  
and  $S$  is thus a l.u.b.  
of  $E$ .